

LESSON 12 / 第十二課

Trees on Built Retaining Structures: From the Coupled System to the Interface

建成擋土結構上的樹木：從耦合系統到界面

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

A tree on a masonry or stone retaining wall is a coupled living–built system. Jim’s stonewall-tree series in humid subtropical cities is the richest published morphology. Failure can start in the stem, at the root–fabric interface, or in the wall. Load sharing is two-way: the wall props the stem; the roots stitch or jack the fabric. A dripline TPZ misses the fabric account. Inspection order starts at the interface. Evidence is rich in description and thin in instrumented load-sharing; transfer to mechanically stabilised earth needs a different fabric account.

磚石或石擋土牆上的樹是耦合的活體–建成系統。Jim 在濕潤亞熱帶城市的石牆樹系列，是已發表形態學中最豐富的。失效可始於莖、根–構件界面或牆本身。荷載分擔是雙向的：牆支撐莖；根縫合或頂起構件。滴水線 TPZ 漏掉構件賬戶。檢查順序從界面開始。描述證據豐富，儀器化荷載分擔證據薄；轉移到加筋土需要另一套構件賬戶。

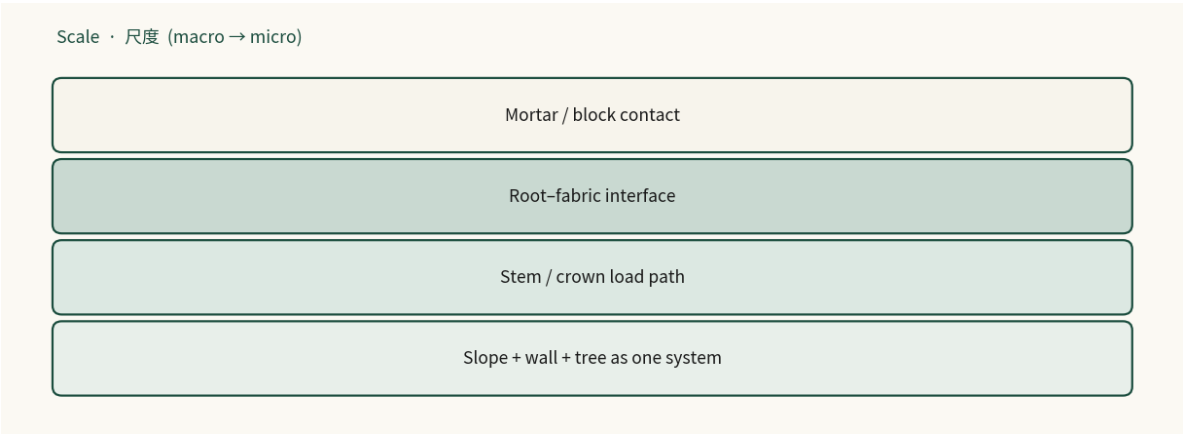


Figure 1. Nested scales from the coupled system to mortar contact.

圖 1. 由耦合系統到砂漿接觸的嵌套尺度。

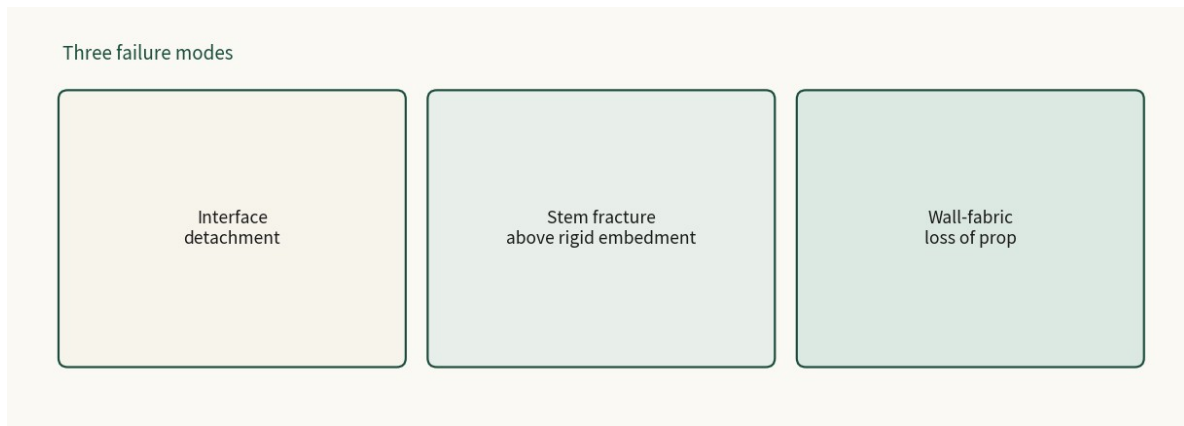


Figure 2. Three recurrent failure modes of wall trees.

圖2. 牆樹反覆出現的三種失效模式。

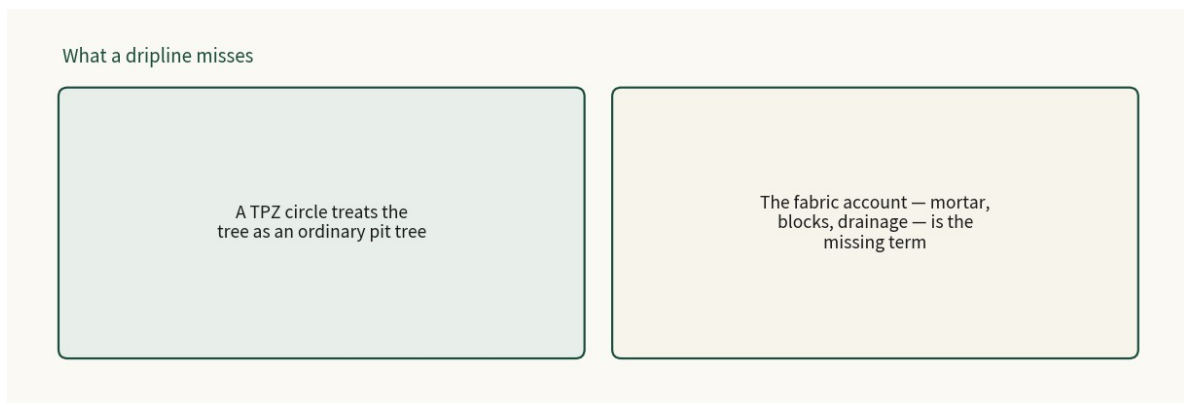


Figure 3. A dripline TPZ misses the fabric account.

圖3. 滴水線 TPZ 漏掉構件賬戶。

1. Scope

1. 範圍

The object is trees structurally entangled with retaining walls. Jim's morphological series supplies the richest published account [1,2].

對象是與擋土牆結構纏結的樹木。Jim 的形態系列提供最豐富的已發表記述 [1,2]。

2. Coupled system and modes

2. 耦合系統與模式

Load sharing is two-way. Aerial roots and buttresses can stitch masonry and can prize it apart [1,2]. Three modes recur: interface detachment, stem fracture above a rigid embedment, and loss of wall fabric until the prop vanishes. Standard mid-stem tomography does not image the interface. Removal of either member changes the other member's residual risk.

荷載分擔是雙向的。氣生根與板根可以縫合砌體，也可以把它撬開 [1,2]。三種模式反覆出現：界面脫離、剛性嵌固上方的莖斷裂，以及牆構件喪失直至支撐消失。標準莖中段斷層不影像化界面。移除任一方都改變另一方的剩餘風險。

3. Inspection order

3. 檢查順序

The first question is the interface: contact, jacking, missing blocks, blocked drainage, decay where root and mortar meet. A dripline TPZ written for an open pit tree does not capture those terms. Tree-risk method still applies — targets, likelihood of a specified part, consequence — but the specified part may be a wall panel.

第一個問題是界面：接觸、頂推、缺塊、排水堵塞、根與砂漿相接處的腐朽。為開敞樹穴樹寫的滴水線 TPZ 捕捉不到這些項。樹木風險方法仍然適用——目標、指定部位的可能性、後果——但指定部位可能是一塊牆板。

4. Limits

4. 限度

Instrumented load-sharing at the interface remains thin. Mechanically stabilised earth, concrete crib walls and soil-nailed slopes are different fabrics. Copying a masonry morphology onto them is a guess.

界面上的儀器化荷載分擔仍然薄。加筋土、混凝土框牆與土釘邊坡是不同構件。把砌體形態抄到它們上面，是猜測。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A wall tree is treated here as:

1. 此處把牆上的樹視為：

- A. An ordinary pit tree.
- A. 普通樹穴樹。
- B. A coupled living–built system.
- B. 耦合的活體–建成系統。
- C. A P50.
- C. 一個 P50。
- D. A register diagnosis.
- D. 名錄診斷。

2. Load sharing is:

2. 荷載分擔是：

- A. One-way, tree only.
- A. 單向，只有樹。
- B. Two-way.
- B. 雙向。
- C. Zero.
- C. 為零。
- D. A 10-20-30 rule.
- D. 10-20-30 規則。

3. Three recurrent modes include:

3. 三種反覆模式包括：

A. Only leaf colour.

A. 只有葉色。

B. Interface detachment, stem fracture, fabric loss.

B. 界面脫離、莖斷裂、構件喪失。

C. Only P88.

C. 只有 P88。

D. Only flush cuts.

D. 只有平切。

4. Mid-stem tomography:

4. 莖中段斷層：

A. Images the interface completely.

A. 完整影像化界面。

B. Does not image the interface.

B. 不影像化界面。

C. Raises mycorrhizae.

C. 提高菌根。

D. Sets occupancy.

D. 設定佔用。

5. A dripline TPZ on a wall tree:

5. 牆樹上的滴水線 TPZ：

A. Captures the fabric account.

A. 捕捉構件賬戶。

B. Misses the fabric account.

B. 漏掉構件賬戶。

C. Equals Wall 4.

C. 等於第 4 壁。

D. Is QTRA.

D. 是 QTRA。

6. Inspection order starts at:

6. 檢查順序始於：

A. Flower scent.

A. 花香。

B. The root-fabric interface.

B. 根-構件界面。

C. i-Tree carbon.

C. i-Tree 碳。

D. A sacred month.

D. 神聖月份。

7. Removing the tree:

7. 移除樹：

A. Leaves the wall's residual risk unchanged.

A. 牆的剩餘風險不變。

B. Changes the wall's residual-risk sentence.

B. 改變牆的剩餘風險句子。

C. Raises P50.

C. 提高 P50。

D. Prints new section.

D. 印出新截面。

8. Transfer to soil-nailed slopes:

8. 轉移到土釘邊坡：

A. Is automatic.

A. 自動的。

B. Needs a different fabric account.

B. 需要另一套構件賬戶。

C. Is forbidden.

C. 被禁止。

D. Equals 3-30-300.

D. 等於 3-30-300。

9. The specified part in the risk sentence may be:

9. 風險句子中的指定部位可以是：

A. Only a leaf.

A. 只有一片葉。

B. A wall panel.

B. 一塊牆板。

C. Only soil pH.

C. 只有土壤 pH。

D. Only a cultivar name.

D. 只有品種名。

10. Instrumented load-sharing evidence is:

10. 儀器化荷載分擔證據是：

A. Thicker than morphology.

A. 厚於形態學。

B. Thinner than morphology.

B. 薄於形態學。

C. Complete.

C. 完整。

D. A cable grade.

D. 鋼纜等級。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Jim C.Y. stonewall-tree morphology series in humid subtropical cities.

[2] Jim C.Y. (2013 and related). stonewall trees as coupled living–built systems.

[3] ANSI A300 Part 9 applied to non-stem parts.

[4] Fite K., Smiley E.T. construction / protection BMPs — limits on dripline rules.

[5] GEO / landscape-treatment literature on walls as engineered fabrics.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。